PATIENT CARE THEORY 2

UNIT 5, PART 5: Complications in Pregnancy

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Patient Care Theory 2
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Topics

- Braxton-Hicks Contractions
- Supine hypotension
- Ectopic pregnancy
- Abortion: Spontaneous; Therapeutic
- PIH; Pre-eclampsia; HELLP Syndrome; Eclampsia
- Gestational Diabetes
- Neuromuscular Disorders
- Placenta Previa
- Abruptio Placenta
- Preterm Labour
- Lack of progress
- Cardiac Arrest

Braxton Hicks Contractions

- Often referred to as “false labour”
- Felt As early as 13 weeks
- Uterus begins intermittent contractions
- May enhance placental blood flow
- Painless (sometimes) irregular contractions
- Do not cause cervical changes
  - not a prehospital assessment

1872

Raines DA, Cooper DB. Braxton Hicks Contractions. [Updated 2019 Nov 11]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2020 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK470546/

Braxton Hicks Contractions

- False labour has the following traits:
  - Irregular in duration and intensity,
  - occur infrequently
  - unpredictable and
  - Decreases with walking
  - May ease with change of position
  - No increase in frequency or intensity
  - More uncomfortable than painful

Braxton Hicks Contractions

- Known triggers can include:
  - Full bladder
  - Activity (exercise)
  - Sexual activity
  - Dehydration

Braxton-Hicks vs True Labour

- True labour includes the following:
  - Become regular
  - Walking increases intensity of contractions
  - Not affected by change of position
  - Frequency and intensity increase

Braxton Hicks Management

- May be difficult to distinguish true from false labour in the field
- Ruling out true labour in the field may place the mom at risk of an unaided premature birth and the paramedic at risk of litigation
- A 911 call is rare and warrants transport in most cases

Braxton Hicks Management

Care in transport?

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Supine hypotension

- Occurs in third trimester
- Gravid uterus compresses inferior vena cava

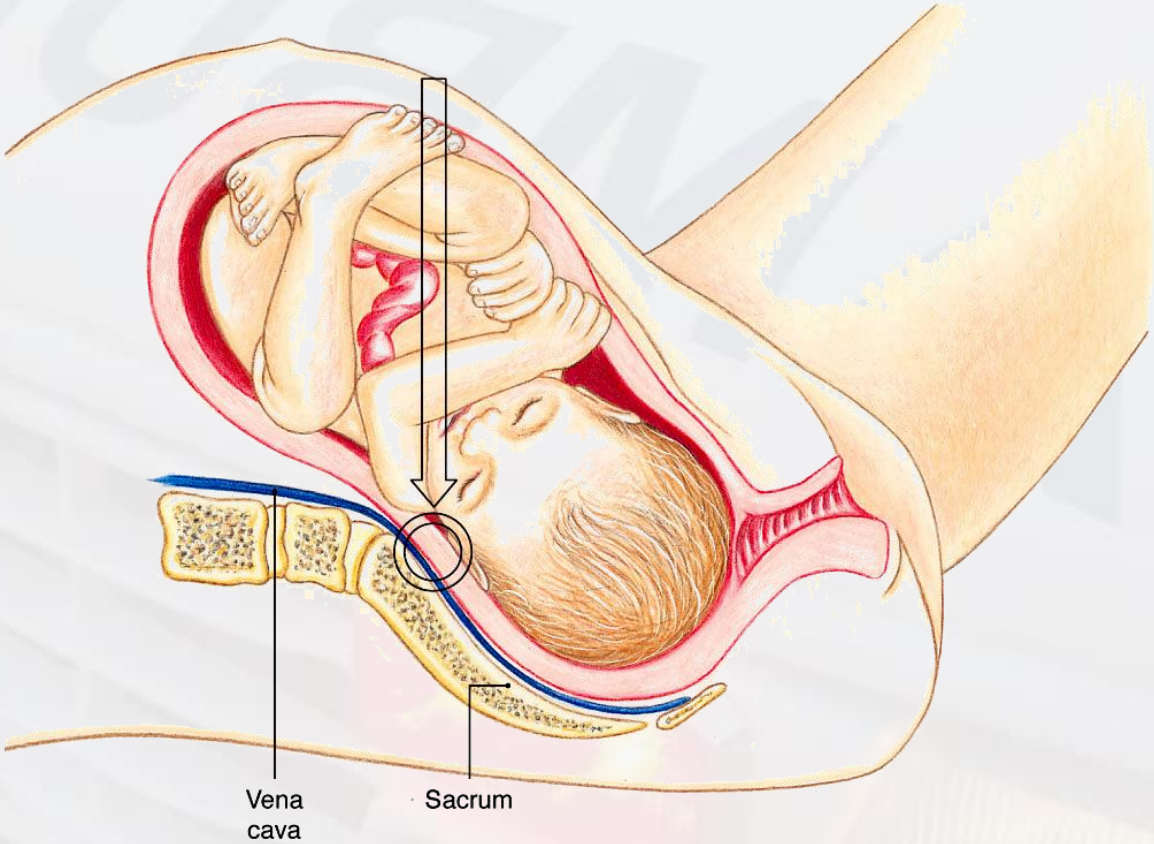


FIGURE 40-7 The supine-hypotensive syndrome results from compression of the inferior vena cava by the gravid uterus.

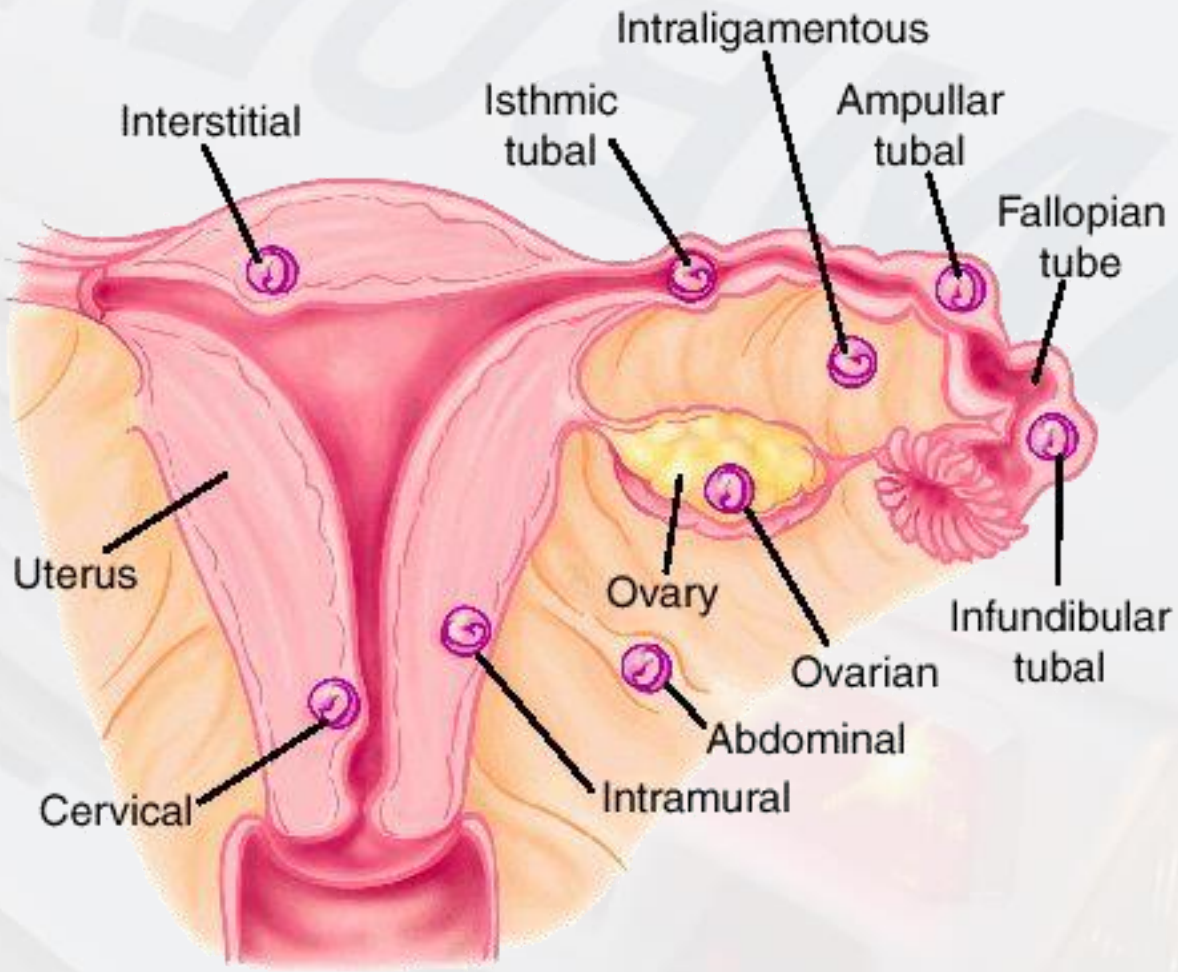
Supine hypotension - Management

- Place mom on stretcher in position of comfort
- If hypotensive, raise right hip 1-15° with rolled blanket
- Monitor maternal vital signs & fetal heart tones (if possible / practical)
- SpO_2
- ECG
- IV prn

Ectopic pregnancy

- flaw in human reproductive physiology that allows the fertilized egg implant and mature outside the uterus
  - e.g. fallopian tubes, cervix, ovary, cornual region of the uterus, and the abdominal cavity
- ends in death of the fetus
- can become a life-threatening situation for the mother
  - leading cause of pregnancy-related death during the first trimester

Ectopic pregnancy



Ectopic pregnancy

Etiology

- anything that hampers the migration of the embryo to the uterine cavity – e.g.
  - Pelvic inflammatory disease, history of prior ectopic pregnancy, history of tubal surgery and conception after tubal ligation, use of fertility drugs or assisted reproduction, increased age, smoking,
- however most patients have no identifiable risk factor

Ectopic pregnancy - Presentation

- classic clinical triad of ectopic pregnancy is
  - pain
  - amenorrhea (absence of menstrual periods)
  - vaginal bleeding
- Unfortunately, only 50% of patients present typically
- Patients may present with other symptoms common to early pregnancy
  - nausea, breast fullness, fatigue, low abdominal pain, heavy cramping, shoulder pain, and recent dyspareunia (pain in the vagina or pelvis experienced during sexual intercourse)

Ectopic pregnancy - Management

- Assume that any female of childbearing age with lower abdominal pain is experiencing an ectopic pregnancy.
- Ectopic pregnancy is life-threatening.
- SpO_2
- O_2 (if bleeding suspected i.e. decreased O_2 carrying capacity)
- ECG
- IV + Fluid resuscitation prn
- Transport the patient immediately

Abortion: Therapeutic/Elective

- Worldwide, 20-30 million legal abortions are performed annually, with another 10-20 million abortions performed illegally
- Abortions performed 1-2 weeks after conception are termed menstrual extractions
- Abortions performed prior to 9 weeks from the last menstrual period (LMP) (7 wk from conception) are performed either surgically or medically.
- 9-14 weeks: performed by a dilatation and suction curettage procedure (D & C)
- After 14 weeks, surgical abortions performed by a dilatation and evacuation procedure

Post-Abortion Complications

- Triad: Pain, bleeding, and low-grade fevers (most common)
  - usually caused by retained products of conception
- Excessive hemorrhage - shock
- Coagulopathy (DIC)
- increased lower midline abdominal pain
- Uterine perforation: increased abdominal pain
- Bowel injury: abd pain, fever, blood in the stool, n/v
- Bladder injury: suprapubic pain and hematuria
- Septic abortion: fever, chills, abdominal pain, and vaginal bleeding
- Failed abortion (continued intrauterine or ectopic pregnancy)

Post-Abortion Management

- Spontaneous or therapeutic abortions have similar risks for hemorrhage
- Treat for shock prn
- SpO_2
- O_2 prn
- ECG
- IV + Fluid resuscitation prn

PIH – Pregnancy Induced HTN

- Develops after 20 wks gestation (in patients with normotension prior to pregnancy)
  - Resolves spontaneously after birth
  - More common in obese women and/or glucose intolerant
  - Can progress to pre-eclampsia
- New terminology: transient hypertension of pregnancy that did not progress to preeclampsia

Pre-eclampsia - HTN

- Pre-eclampsia is the most common and most serious hypertensive disorder (50-70%)
- Also referred to as “toxemia of pregnancy”
  - Terms used: “the patient is toxemic” or “the patient is pre-eclamptic”
  - Risk factors: <18, >35, chronic HTN, renal disease, diabetes, race (African-Canadian most susceptible)
- Etiology unknown
  - Hypothesis: placental dysfunction may initiate the systemic vasospasm, ischemia, and thrombosis –that eventually damages maternal organs (eMedicine: http://www.emedicine.com/emerg/topic480.htm)

Pre-eclampsia - HTN

- Occurs after 20 weeks gestation
  - **Unless a pre-existing chronic HTN exists**
- one of the most common causes of perinatal morbidity and mortality
- Characterized by gestational HTN + one of the following:
  - New proteinuria; edema (feet/ankles, face, hands)
- HELLP syndrome – life threatening liver disorder related to pre-eclampsia (a severe form of pre-eclampsia)
  - **H**emolysis
  - **E**levated **L**iver enzymes
  - **L**ow **P**latelet count

Pre-eclampsia - HTN

- Mild Pre-eclamsia
  - ↑ in systolic BP of 30 mmHg + ↑ in diastolic BP of 15 mmHg,
  - or an absolute reading of >140 mmHg/90 mmHg
  - Mild peripheral edema
  - Headaches
  - Proteinurea (urine dipped for protein in the hospital)

Pre-eclampsia - HTN

- Severe Pre-eclampsia
  - SBP > 160 mmHg or DBP > 105 mmHg can cause end-organ damage
  - Persistent/severe frontal headache
  - visual disturbances
  - confusion
  - Upper abdominal pain
  - Nausea/vomiting
  - oliguria
  - peripheral edema (hands, feet/ankles, face)
  - pulmonary edema

Pre-eclampsia - Management

Prehospital care

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Eclampsia

- Defined by: **seizures / Coma**
  - Pre-eclampsia may lead to eclampsia
  - Likely due to hypertensive encephalopathy
- Occurs in 0.2% of pregnancies
- Life-threatening!
  - Intracranial hemorrhage is the major cause of maternal death
    - maternal death = fetal death

Eclampsia - Management

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- Transport!!

- Cure = birth

Gestational Diabetes

- Affect between 3 and 20% of pregnant women
- Hormonal influences and increased tissue response
- High risk for later development of diabetes
- Consider hypoglycemia in patient with decreased LOC
- Administer dextrose as indicated
  - Glucagon to be administered if D50W is not an option

Gestational Diabetes

- Can lead to pre-eclampsia
- Abnormal sugar levels in the newborn (low)
  - What is the normal newborn blood sugar?
- Large babies – difficulty with normal vaginal delivery (increases the need for C-section)

Neuromuscular Disorders

Myasthenia Gravis

- Autoimmune disorder in which acetylcholine receptors are destroyed
- Leads to skeletal muscle weakness – possibly respiratory distress/failure
  - commonly affected muscles are those of the eyes, face, and swallowing, as well as those responsible for breathing
  - Hallmark presentation is muscle weakness that worsens after periods of activity and improves with rest

Neuromuscular Disorders

Myasthenia Gravis

- Occurs in 1 out of every 20,000-40,000 pregnancies **(eMedicine: http://www.emedicine.com/med/topic3260.htm)**
- Antepartum exacerbations are common: 41% in one study **(eMedicine: http://www.emedicine.com/med/topic3260.htm)**
- Many treatment options available
  - Thymectomy
  - Anticholinesterase medications
  - Immunosuppressive drugs

Neuromuscular Disorders

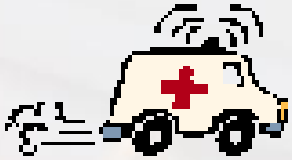
Assessment-Management of Myasthenia Gravis

- Respiratory assessment is critical
  - SOB? Dyspnea?
  - Assess RR and T_V, SpO_2, LOA
- Patient may not have the strength to push to deliver the baby → transport may be critical

Neuromuscular Disorders

Management

- SpO_2
- ECG
- O_2 prn
- PPV prn
- left lateral recumbent position prn
- IV TKO
- Transport



Placenta Previa

- Abnormal implantation of placenta on the lower half or the uterus
  - The placenta is implanted near the cervical opening and may partially or completely cover it
- Painless vaginal bleeding (3rd trimester)
- Never attempt vaginal exam
- Treat for shock
- Rapid transport
- Neonatal risks - preterm birth, RDS, low APGAR scores, anemia

Placenta Previa

Total placenta previa

Partial placenta previa

Placenta Previa - Management

Treat for shock

- SpO_2
- ECG
- O_2 PRN
- left lateral recumbent position prn
- IV
- Fluid resuscitation prn
  - Placenta previa is thought to account for 31% of all antepartum hemorrhage
- Rapid transport

Definitive management is delivery by c-section

Abruptio Placenta/ Placental abruption

- Premature separation of a normally implanted placenta from the uterus: Classified as partial, severe, or complete
- Bleeding may be apparent (vag bleeding) or may be concealed behind the placenta
- Frequency: 1% of all pregnancies
- Occurs most commonly in 3rd trimester (but can occur before)
- Life threat for mother and fetus - ~22% of antepartum hemorrhage and leading cause of vaginal bleeding in the 2nd half of pregnancy
  - Fetal mortality ~15%, perinatal mortality 9-30%
- Maternal and fetal death may occur because of hemorrhage and coagulopathy.

Abruptio placenta

- Risk Factors
  - Maternal hypertension (most common cause) occurring in approximately 44% of all cases
  - Advanced maternal age
  - Idiopathic
  - Previous placental abruption
  - Smoking/ alcohol/ cocaine during pregnancy
  - Trauma - causes 1.5-9.4% of all cases
  - Premature rupture of membranes (24hrs or more)
    - Risk increases when infection present
  - Short umbilical cord

Abruptio Placenta

- S&S include
  - +\- Vaginal bleeding (blood can be trapped inside the uterus)
  - Abdominal pain (sudden onset)
  - Back pain
  - Uterine tenderness or rigidity
  - Continuous uterine contractions
  - Decreased fetal movement

Abruptio Placenta

Partial separation (concealed hemorrhage) | Partial separation (apparent hemorrhage) | Complete separation (concealed hemorrhage)

Abruptio Placenta - Management

- Treat for shock
- left lateral recumbent position prn
- SpO_2
- O_2 prn
- ECG
- IV
- Fluid resuscitation prn

Preterm Labour

- presence of uterine contractions of sufficient frequency and intensity to effect progressive effacement and dilation of the cervix prior to term gestation (between 20 and 37 wk).
- occurs in approximately 12% of pregnancies
- leading cause of neonatal mortality

Preterm Labour

- Preterm premature rupturing of the membranes (PPROM) is associated with 30-40% of preterm deliveries and is the leading identifiable cause of preterm delivery.
- Risk factors for PPROM and preterm labour
  - smoking, previous preterm delivery, or vaginal bleeding at any time during the pregnancy

- **Note: "premature rupture of membranes" or PROM occurs at ≥ 38 weeks but before the onset of labour**

Preterm Labour

- Precipitating factors
  - Infection
    - e.g. vaginosis, STDs, UTIs, etc
  - uterine distention – e.g. multiple gestation, polyhydramnios (too much amniotic fluid)
  - uterine distortion
  - incompetent cervix
  - abruptio placentae
  - placenta previa
  - HTN
  - IDDM

Preterm Labour

- Precipitating factors
  - Cardiovascular disease
  - Renal disease
  - Past history of preterm labour
  - multiparous
  - Trauma

Preterm Labour

- Precipitating factors
  - drug abuse
  - smoking
  - alcohol consumption
  - stressful life situations (e.g. domestic violence; close family death; insecurity over food, home, or partner)
  - Recently, the role of the fetus in the initiation of labor has been recognized. In a simplistic sense, the fetus recognizes a hostile intrauterine environment and precipitates labor.
    - Future tocolytic therapy may be directed at modifying fetal response.

Preterm Labour - Management

- SpO_2
- O_2 prn
- ECG
- left lateral recumbent position prn
- IV
- Rapid transport to avoid delivery in the field (if possible)
- Be prepared for a precipitous delivery

Lack of Progress of labour

- Situations where signs of imminent birth are present but labour fails to progress to delivery within ~ 10 min
  - Primips
  - Baby is “face up” (occiptoposterior)
  - Baby is positioned facing patient’s anterior instead of posterior
- Management:
  - IF birth has not occurred within 10 minutes of initial assessment - transport

Pregnancy & Cardiac Arrest

- Goal ROSC in Mom
- Initiate transport ASAP
- C-SECTION should be performed in hospital within 4 minutes of an ROSC (this may be impossible when the arrest is outside the hospital)

QUESTIONS